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Oxidizing Impact Induced by Mackinawite (FeS) Nanoparticles at Oxic Condition due to Production of Hydroxyl Radicals

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ABSTRACT Mackinawite (FeS) nanoparticles have been extensively tested for reducing contaminants under anoxic conditions, while the oxidizing impact induced by FeS under oxic conditions has been largely underestimated. In light of previous finding that hydroxyl radicals (•OH) can be produced from oxygenation of sediment Fe(II), herein we revealed that •OH can be produced efficiently from FeS oxygenation at circumneutral conditions, yielding 84.7 μmol •OH per g FeS. Much more •OH was produced from the oxygenation of FeS compared with siderite, pyrite and zerovalent iron nanoparticles under the same conditions. The oxidation of FeS was a surface-mediated process, in which O_2 was transformed by the structural Fe(II) on FeS surface to •OH with the generation of H_2O_2 intermediate. A small proportion of Fe(II) was regenerated from the reduction of Fe(III) by FeS and S(-II), but this proportion did not significantly contribute to •OH production. We further validated that the •OH produced from FeS oxygenation considerably contributed to the oxidation of arsenic. As the change of redox condition from anoxic to oxic is common in both natural and artificial processes, our findings suggest that the oxidizing impact induced by FeS at oxic condition should be concerned due to •OH production.

INTRODUCTION

Mackinawite (FeS) nanoparticles are widespread in subsurface anoxic environments due to the biological reduction of sulfate.[1] FeS engineering nanoparticles have been applied for environmental remediation.[2] In both natural and artificial processes, there are a large number of investigations regarding the ability of

FeS in transforming and removing contaminants under anoxic conditions.[2] The most striking property of FeS is its superior reducing ability to transform halogenated organic compounds,[3] Cr(VI),[4] U(VI),[5] Tc(VII),[6] etc. Due to the chalcophilic nature, FeS can capture many divalent metals[7,8] and arsenic[9] through forming surface complexes, insoluble metal sulfides or isomorphous substitution.

However, there are limited investigations reporting the oxidative transformation of contaminants induced by FeS under oxic conditions. Due to the high susceptibility of FeS to oxidation by O_2, FeS is considered as a redox buffer inhibiting the oxidative remobilization of inorganic contaminants like U[10] and Tc[11] when the redox condition changes from anoxic to oxic. Virtually, the oxidizing impact induced by FeS under oxic conditions has been noted or hidden in several studies. For example, the concurrent oxidation of As(III) with the oxygenation of FeS was noted by Hayes's group, which was presumably attributed to the production of reactive oxidants such hydroxyl radicals (•OH) and Fe(IV).[12,13] Recently, the oxidative mobilization of noncrystalline U(IV) coupled with FeS oxygenation was further reported by the same group, and the reactive oxidant was supposed to be a transient surface Fe(III) species.[14] Another recent study presented the hidden oxidative mobilization of Tc(IV) sequestrated on sulfidated nano zerovalent iron (nZVI) at a low S/Fe ratio or in the latter stage of oxygenation.[11] It is therefore concluded that FeS oxygenation could induce the oxidative transformation of contaminants at certain conditions, but the reactive oxidant accounting for the oxidation is not clear.

For the transformation and treatment of contaminants, many Fe(II)-containing

58 substances such as ligand-complexed Fe(II), siderite, magnetite and pyrite with
59 discrepant reducing abilities have shown the potential to activate molecular O_2 for
60 •OH production.[15–22] Due to the extremely high oxidation ability (standard reduction
61 potential: 2.8 V[23]), •OH can oxidize most organic contaminants and redox-sensitive
62 elements at near diffusion-controlled rates.[24,25] For example, •OH resulting from the
63 oxidation of ligand-complexed Fe(II) by O_2 efficiently degraded organic pollutants in
64 wastewaters,[15,16] the transformation of siderite to goethite upon oxygenation
65 concurrently oxidized As(III) to As(V) because of possible •OH production,[17]
66 interactions of magnetite with O_2 oxidized As(III) and nalidixic acid at neutral pH by
67 means of Fenton-like reactions,[18,19] and •OH was measured from the oxidation of
68 pyrite by O_2 under acidic conditions[20] leading to the oxidation of trichloroethylene
69 and Sb(III).[21,22] Our recent study highlighted the importance of Fe(II) minerals in
70 subsurface sediments for •OH production under oxic conditions.[26] However, it is not
71 clear whether oxygenation of FeS could produce •OH.

72 In light of the concurrent oxidizing impact coupled with FeS oxygenation[12,14] and
73 the high efficiency of structural Fe(II) in minerals for •OH production,[26] we
74 hypothesize that •OH can be produced upon FeS oxygenation and contributes to
75 contaminant oxidation. In this study, we quantitatively measured the cumulative •OH
76 produced from FeS oxygenation employing the transformation of benzoate (BA) to
77 para-hydroxybenzoic acid (*p*-HBA) as a probe reaction.[27] The mechanisms of •OH
78 production were unraveled through exploring FeS oxidation process and O_2 reduction
79 pathway. The contribution of •OH produced from FeS oxygenation was ultimately

evaluated for the concurrent oxidation of As(III). This study aims to supplement the fundamentals for the oxidizing impact induced by FeS on contaminant transformation under oxic conditions.

EXPERIMENTAL SECTION

Chemicals. Sodium benzoate (BA, 99.5%), *p*-HBA (99%) and 2, 2′-bipyridine (BPY, 99.5%) were purchased from Sinopharm Chemical Reagent Co., Ltd, China. Sodium sulfide, nonahydrate ($Na_2S•9H_2O$, 98.0%) was purchased from Shanghai Tongya Chemical Technology Co., Ltd. Arsenic (As_2O_3, 99.8%) was purchased from Shanghai General Reagent Factory, China. $Na_2HAsO_4•7H_2O$ (99.99%) and 5,5-Dimethyl-1-99 pyrroline-N-oxide (DMPO) were obtained from Sigma-Aldrich. Piperazine-N,N-bis (ethanesulfonic acid) (PIPES, Aladdin Chemistry Co. Ltd., China) was used as the buffer because it does not form complexes with Fe(II) or Fe(III).[28] Nitro blue tetrazolium (NBT) was of analytical grade (Amresco, America). Superoxide dismutase (SOD) was obtained from Shanghai Kayon Biological Technology Co., Ltd. Deionized (DI) water (18.2 MΩ·cm) from a Heal Force NW ultra-pure water system was used for all experiments and the other chemicals used were analytical reagents of high purity.

Preparation of FeS nanoparticles is referred to the procedure reported by Butler and Hayes.[29] A total of 72 mL of 1.1 M Na_2S was slowly added to 120 mL of 0.57 M $FeCl_2$ in an anoxic glove box (Mikrouna, China) filled with ultrapure Ar gas (99.999%). The resulting suspensions were mixed for 3 days and then transferred into

polypropylene bottles that were tightly sealed. The bottles were moved out from the glove box and centrifuged at 10000 rpm for 10 min. The supernatant was replaced with fresh deoxygenated water in the glove box, and the bottles were centrifuged again. This procedure was repeated for several times until the conductivity was below 200 μs/cm. The resulting particles were identified to be mackinawite by X-ray diffraction (XRD, Figure S1 in the Supporting Information (SI)) and X-ray photoelectron spectroscopy (XPS, SI Figure S2). The average hydrodynamic diameter and zeta potential of stock FeS suspension (pH 7) were determined to be 100~200 nm and -32.51 mV, respectively, using the dynamic light scattering (DLS, Zetasizer, Malvern). FeS stock suspension was stored in the anoxic glovebox for later use.

Oxygenation of FeS. All the oxygenation experiments were conducted in a stirred reactor (100 mL) containing 50 mL reactant solution with 10 mM BA at room temperature (25 ± 1°C) in the dark. The reactor was enwrapped with silver foil to avoid any potential photochemical reactions of FeS that could generate reactive oxygen species. The reactor was open to the atmosphere through several small pores in the top, and the variation of dissolved oxygen (DO) concentration was measured by a DO probe (JPB-607A, Shanghai INESA) in the reactor. The suspension pH was buffered at 7.0 (± 0.5) by 3 mM PIPES in all the experiments. The reaction rate constant for PIPES and •OH is not available but can be supposed to be smaller than that for BA and •OH ($k_{BA,\ \cdot OH}$ = 5.7 × 10^9 M^{-1} s^{-1} [30]) which is nearly diffusion-controlled. So, the competition of PIPES with BA on the scavenging of •OH could be of minor importance under the experimental conditions. The oxidation was

initiated by adding specific volumes of the stock FeS suspension to produce the dosages of 0.1, 0.5, 1 and 3 g/L, respectively. The quenching experiments with additions of 0.5 mM BPY, 1 mM NBT and 60 U/L SOD were carried out in the same reactor. Anoxic control experiments were conducted in the anoxic glove box. At predetermined time intervals (0–4 h), about 4 mL of suspension were withdrawn, centrifugzed at 10000 rpm for 1 min, and filtered immediately through a 0.22-μm nylon filter. The filtrate was analyzed for *p*-HBA, H_2O_2, dissolved Fe^{2+}, $S_2O_3^{2-}$ and SO_4^{2-} concentrations. All the experiments were carried out at least in duplicate.

Oxygenation of Different Forms of Reduced Iron. Oxygenation of different forms of reduced iron was compared using siderite ($FeCO_3$), pyrite (FeS_2) and nZVI because all of them have been reported to produce •OH under oxic conditions.[20,22,26,31] Preparation of siderite and pyrite can be referred to our previous work.[26,32] nZVI was prepared as described by Lee et al.[33] The as-prepared particles were dried in an anoxic glove box. The specific surface areas of the FeS, FeS_2 and nZVI particles were measured to be 42.3, 3.5 and 5.0 m^2/g, respectively, through a multipoint BET (Brunauer, Emmett, and Teller) analysis with N_2 adsorption at 77 K on a Micromeritics surface area analyzer (ASAP-2020). The specific surface area of $FeCO_3$ was not obtained because of its fast oxidation during the measurement. The content of Fe in all the iron materials was set the same at 0.5 g/L for comparison (that is, 0.79 g/L FeS, 1.04 g/L $FeCO_3$, 1.08 g/L FeS_2 and 0.50 g/L nZVI). The oxygenation experiments were carried out under identical conditions as mentioned above.

Contaminant Oxidation by the •OH Produced from FeS Oxygenation.

Because arsenic contamination in groundwater is serious in the world[34] and As(III) oxidation coupled with FeS oxygenation has been noted previously,[12] we examined the contribution of •OH produced from FeS oxygenation to As(III) oxidation. The oxidation was performed by adding 1000 μg/L As(III) into the aforementioned oxygenation reactor containing 1 g/L FeS suspension. Because organic buffers such as PIPES could scavenge •OH screening the oxidation of As(III) caused by •OH, the initial suspension pH was adjusted to 7 by dilute H_2SO_4. The variation of pH was measured to range at 6.5–7.0 during the process. To evaluate the oxidation induced by •OH, quenching experiments were carried out by respectively adding 100 mM methanol and 10 mM BA.

Analysis. The concentration of *p*-HBA was measured in an LC-15C HPLC (Shimadzu) equipped with a UV detector and an Inter Sustain C18 column (4.6 × 250 mm). The mobile phase was a mixture of 0.1% trifluoroacetic acid aqueous solution and acetonitrile (65:35, v/v) at a flow rate of 1 mL/min, with the detection wavelength at 255 nm. A conversion factor of 5.87 was used to estimate the cumulative •OH concentrations.[26] Note that the concentrations of •OH presented in this study denoted the aqueous •OH which can be trapped by BA. There could be a small portion of •OH adsorbed on solid (i.e., FeS) surface, but quantifying this portion is currently difficult.

H_2O_2 was analyzed by a modified DPD method at 551 nm using a UV-vis spectrophotometer (UV-1800 PC, Shanghai Mapada Spectrum Instrument Co., LTD).[35] During the measurement, BPY and Na_2EDTA were added to complex Fe^{2+} and Fe^{3+}, respectively. For the measurement of total iron, the samples were

completely dissolved by 6 M HCl. Fe^{2+} was measured by the 1, 10-o-phenanthroline analytical method at 510 nm. The concentration of elemental sulfur was measured by an LC-15C HPLC. $S_2O_3^{2-}$ and SO_4^{2-} was measured by an ionic chromatograph equipped with a suppressed conductivity detector (Metrohm 761 compact IC), a Metrosep A Supp 4 analytical column (250 × 4.0 mm) and a Metrosep A Supp 4/5 guard column. As(III) and As(V) in filtered samples were measured on an HPLC coupled to an atomic fluorescence spectrometer (AFS 9600, Beijing Kechuang Haiguang Instrument Co., Ltd).[36] Unfiltered samples were dissolved by 6 M HCl for the analysis of total As(III) and As(V). When S(-II) were contained in the sample, its influence on As(III) analysis was eliminated by the addition of 40 mM NaOCl[37] in another set of samples. In this case, As(III) concentration was obtained from the difference of total As and As(V).

Characterization. At predetermined time intervals, the suspension from the reactor was purged with N_2 to remove O_2, centrifuged at 10000 rpm, and then dried in the anoxic glove box. XRD patterns were obtained on a D8-FOCUS X-ray diffractometer with Cu K radiation (Bruker AXS., Germany) at 40 kV and 40 mA and at the scanning step size of 0.01^0 and step time of 0.05 s. Qualitative identification of mineral phases was made using the MDI Jade 5.0 software. XPS spectra was performed on a VG Multilab 2000 X-ray Electron Energy Spectrometer (Thermo Fisher Scientific, USA) using a monochromatic Al Kα radiation (Power 300 W) and a low-energy electron flooding for charge compensation. High resolution spectra were fitted using a least-square procedure with a Gaussian-Lorentzian peak shape after

subtracting a Smart baseline (Avantage 4.88). Fe K-edge extended X-ray absorption fine structure (EXAFS) spectroscopy was collected at the beamline 1W1B at the Beijing Synchrotron Radiation Facility (BSRF). The electron beam energy of BSRF was 2.5 GeV with a maximum beam current of 250 mA. The monochromator energy was calibrated using an Fe foil before every sample run. Spectra were energy calibrated with the software Average and normalized and background corrected with standard features of the ATHENA software package.[38] Linear combination fits were carried out in ATHENA using lepidocrocite and pristine FeS references as standards based on XRD analysis and previous report.[12]

RESULTS AND DISCUSSION

•OH Production from FeS Oxygenation. Using the hydroxylation of BA to *p*-HBA as a probe reaction,[27] the cumulative concentrations of •OH produced from the oxygenation of 1 g/L FeS was quantified. The cumulative •OH gradually increased to a plateau concentration of 115.7 μM within 3 h (Figure 1). However, the cumulative concentration was below the detection limit at anoxic condition, and only attained 4.81 μM with the addition of 1 M ethanol, an •OH scavenger.[31] Control experiments with respective addition of $HgCl_2$ and NaN_3 precluded the significant contribution of biological process (data not shown). Our recent work also proved that microbial process negligibly contributed to •OH production upon the oxygenation of sediments taken from the field.[27] We therefore conclude that •OH was produced from the chemical oxidation of FeS by O_2.

In comparison, at the same content of iron (0.5 g/L), the cumulative concentrations of •OH produced from oxygenation of different forms of reduced iron were quite different, being 53.1 μM for FeS, 20.2 μM for $FeCO_3$, 1.2 μM for FeS_2 and 1.1 μM for nano Fe^0 (SI Figure S3a). Although all the tested reduced iron was reported to produce •OH at oxic conditions,[20,22,26,31] the production efficiency from FeS oxygenation is considerably higher. The specific surface area of FeS (42.3 m^2/g) is only 12.1 and 8.5 times higher than those of FeS_2 (3.5 m^2/g) and Fe^0 (5.0 m^2/g), respectively, so the higher specific surface area was not the main reason for the higher concentration of •OH from FeS oxygenation. Virtually, during the course of oxygenation, FeS and $FeCO_3$ were almost completely oxidized within 4 h (SI Figure S3b), while pyrite and nZVI were apparently oxidized to a very small extent. Therefore, the different reactivity of the tested iron materials is supposed to control the efficiency of •OH production upon oxygenation, while the exact mechanism needs further elucidation.

Mechanisms of •OH Production. *(a) FeS Oxidation.* To explore the dependence of •OH production on FeS oxidation, we measured the cumulative concentrations of •OH produced at different dosages of FeS. With the increase in the dosage from 0.1 to 0.5, 1 and 3 g/L, the cumulative •OH within 4 h increased accordingly from 13.7 to 61.3, 117.2 and 266.0 μM (Figure 2a). The cumulative concentrations of •OH within 4 h are linearly dependent on the dosages of FeS (R^2 = 0.98, SI Figure S4). The slope is 84.7, indicating the production of 84.7 μmol •OH from the complete oxidation of 1 g (or 11.4 mmol) FeS. This relation agrees with previous finding that the cumulative

concentrations of •OH produced from sediment oxygenation linearly depended on sediment Fe(II) content.[26]

A lag time for •OH production was observed in the initial stage (Figure 2a), which apparently increased with the increase in FeS dosage. After the lag time, •OH production increased dramatically. This trend is roughly consistent with the decrease in total Fe(II) concentration (Figure 2b). Note that Fe(II) mainly existed in solid form because the dissolved concentrations were always lower than 0.08 mM (SI Figure S5a). The DO concentration in the initial stage decreased with the increase in FeS dosage and augmented with the progress of oxygenation (SI Figure S6). Particularly, the DO concentration was negligible within the initial 1 h upon oxygenation of 3 g/L FeS. In this regard, high dosages of FeS are used as redox buffer to scavenge O_2.[10,11] Differently, the production of sulfur compounds, predominantly elemental sulfur, was fast initially (Figure 2c, SI Figure S5b). Moreover, the elemental sulfur produced were much more than the total Fe(II) oxidized in the initial stage (SI Table S1). For the abiotic oxidation of FeS by O_2 at neutral pH, it is believed that structural Fe(II) and S(-II) undergo independent oxidations and Fe(II) is oxidized prior to S(-II).[11,12,39] This mechanism was partly supported by the formation of surface Fe(III)-S and Fe(III)-O in Fe $2p_{3/2}$ XPS spectra (SI Figure S2). Since the molar ratio of S to Fe is 1:1 in FeS, there must be a hidden source feeding Fe(II) which apparently alleviated the decrease in total Fe(II) in the initial stage lacking O_2. For the oxidation of sulfide by O_2, the catalytic role of iron has been well recognized.[1,40] That is, Fe(II) is quickly oxidized by O_2 to Fe(III) at neutral conditions, which subsequently oxidizes sulfide with

regeneration of Fe(II).[40–43] Thus, it is proposed herein that FeS was oxidized by O_2 producing Fe(III), which was in turn reduced by sulfide to Fe(II). This pathway is supposed to be the hidden source feeding Fe(II) during FeS oxygenation. As a consequence, both the structural Fe(II) in FeS and the Fe(II) fed from Fe(III) reduction were oxidized by O_2, probably co-contributing to •OH production.

A close correlation between the total Fe(II) and cumulative •OH was obtained for all the dosages tested ($R^2 > 0.97$, Figure 2d). This correlation suggests that •OH was produced from the net oxidation of total Fe(II), regardless of the regeneration of Fe(II) from Fe(III) reduction. The Fe(II) fed from Fe(III) reduction by sulfide involved the free Fe^{2+} and the hydrolyzed and adsorbed forms. As oxygenation of these forms of Fe(II) at neutral pH does not significantly produce •OH,[26,35] it is rational to speculate that •OH was mainly produced from the oxidation of structural Fe(II) in FeS. The slopes of correlation between the total Fe(II) and cumulative •OH are nearly constant (-0.8 ± 0.2) at different dosages of FeS (Figure 2d), further supporting that •OH production was independent of the regeneration of Fe(II) from Fe(III) reduction. The initial lag time for •OH production can be ascribed to the competition of high concentrations of Fe(II) and sulfur with BA for scavenging •OH.

XRD patterns presented that lepidocrocite and elemental sulfur were produced during the oxygenation of FeS (Figure 3a). Fe K-edge EXAFS analysis also showed the production of lepidocrocite mainly (Figure 3b). The EXAFS results revealed that FeS was oxidized by 49.0% at 2 h and by 97.6% at 4 h for the oxygenation of 3 g/L FeS. The total Fe(II) concentration calculated from FeS content (51.0%) was 17.4

mM at 2 h, which was slightly lower than the total Fe(II) concentration measured (21.0 mM). This slight difference agreed with the regeneration of Fe(II) from Fe(III) reduction. Whereas, the slight difference also suggested that the proportion of Fe(II) fed by Fe(III) reduction was low, i.e., about 17.3%. This proportion of Fe(II) was not characterized by XRD and EXAFS probably because of the poor crystalline and quick oxidation by O_2. For example, a green rust-like phase has been previously measured during the oxygenation of FeS at pH 7.1.[12] Production of lepidocrocite and elemental sulfur is consistent with the surface-mediated oxidation of FeS at neutral pH.[12]

It is therefore concluded that •OH was mainly produced from the oxidation of structural Fe(II) in FeS by O_2, although the Fe(II) fed by Fe(III) reduction was also oxidized. FeS oxidation at neutral pH was predominantly surface-mediated, which mainly produced lepidocrocite and elemental sulfur through Eq. 1.

$$FeS(s) + 3/4\ O_2 + 1/2\ H_2O = 1/8\ S_8(s) + \gamma\text{-}FeOOH\ (s) \quad (1)$$

(b) O_2 Reduction. In addition to FeS oxidation, O_2 reduction is the other half reaction constituting FeS oxygenation. The oxygenation of Fe(II) generally conforms to the Haber-Weiss mechanism.[44] That is, Fe(II) donates electrons to O_2 producing $O_2^{\bullet-}$ or H_2O_2, which is further decomposed to •OH (or other reactive oxidants) by Fe(II). To explore the evolution of O_2 to •OH during FeS oxygenation, we first measured the instantaneous concentration of H_2O_2 produced from FeS oxygenation. For all the dosages of FeS oxygenated, H_2O_2 concentration increased rapidly in the initial 0.5 h and decreased to below the detection limit at 2 h (Figure 4a). The peak concentration of H_2O_2 was higher for the oxygenation of higher dosages of FeS. H_2O_2

concentrations were always below the detection limit in the anoxic FeS suspension. The time profile of H_2O_2 production was similar to that from pyrite oxygenation.[32] In the initial stage, the high concentrations of Fe(II) donated electrons to O_2, leading to the pronounced production of H_2O_2. With the progressive oxidation of Fe(II), the production rate of H_2O_2 decreased and was gradually outcompeted by the consumption rate, resulting in the decrease in H_2O_2 accumulation.

During the course of FeS oxygenation, H_2O_2 can be produced from O_2 through oxidation of the structural Fe(II) in FeS, the Fe(II) fed by Fe(III) reduction and the low concentrations of dissolved Fe(II). As BPY has a strong chelating ability with Fe^{2+},[35] it can quickly chelate and deactivate the dissolved Fe(II), thereby screening its oxidation by O_2. However, the addition of BPY into the oxic FeS suspension did not cause any significant change of H_2O_2 production (Figures 4b). The concentration of BPY added (0.5 mM) was much higher than the dissolved Fe(II) (< 0.05 mM, SI Figure S5a). Thus, the dissolved Fe(II) contributed negligibly. Although production of H_2O_2 from oxidation of Fe(II) resulting from Fe(III) reduction by sulfide has been experimentally validated,[45] the contribution of the portion of fed Fe(II) could be less important than the structural Fe(II) in FeS. The appearance of H_2O_2 has been measured electrochemically in the field with co-existence of O_2, Fe(II) (including FeS) and sulfide,[46,47] which partly supports the production of H_2O_2 from FeS oxygenation.

In order to probe the number of electrons transferred from Fe(II) to O_2 for H_2O_2 production, the formation of $O_2^{\bullet-}$, a one-electron transfer intermediate, was examined.

With the addition of 0.5 mM NBT, an $O_2^{\bullet -}$ scavenger,[48] negligible influence on H_2O_2 or •OH production was observed (Figures 4b, 4c). Similarly, the influence of SOD was also negligible. As a result, a two-electron transfer process was supposed to predominate for the reduction of O_2 to H_2O_2 during FeS oxygenation. As the oxygenation of the Fe(II) fed by Fe(III) reduction proceeds through the one-electron transfer process,[49,50] the two-electron transfer process applied for the oxidation of FeS by O_2. One-electron transfer process has been widely reported for the oxidation of Fe^{2+},[44,51] adsorbed Fe(II)[15,49,50] and sediment Fe(II)[26] according to the Haber-Weiss mechanism. The two-electron transfer process has been noted for reduction of O_2 to H_2O_2 by pyrite[32] and ZVI.[27,49,50] FeS possesses a special tetragonal layer structure, and the short Fe-Fe distance (0.26 nm) within the layers leads to the metallic conducting property because of extreme delocalization of the *d* electrons in the basal plane.[52] In addition, the overlap of *3d* orbitals among the neighboring Fe atoms in the energy band structure of FeS gives rise to the highest electron-occupied orbital.[12] Consequently, two electrons are expected to be transferred simultaneously among the neighboring Fe atoms during the surface-mediated oxidation process.

Likewise, H_2O_2 can be decomposed to •OH by the structural Fe(II) in FeS, the Fe(II) fed by Fe(III) reduction and the dissolved Fe(II) during the oxygenation process. Similarly, BPY was added to screen the decomposition by the dissolved Fe(II) (Figure 4c). The minimal influence of BPY precluded the contribution of dissolved Fe(II). To evaluate the role of FeS in decomposing H_2O_2 into •OH, 1 mM H_2O_2 was mixed with 1 g/L FeS at pH 7 in the anoxic glove box. The cumulative •OH rapidly increased to

8.92 μM within 0.5 h and stabilized later on (SI Figure S7), proving the effectiveness of FeS in decomposing H_2O_2 to •OH. Due to the fact that Fe(II) oxidation by H_2O_2 at neutral pH does not significantly produce •OH,[35] FeS predominated for the decomposition of H_2O_2 to •OH. This conclusion was consistent with the aforementioned result that •OH production was mainly due to the net oxidation of total Fe(II).

Contribution of the •OH to As(III) Oxidation. Efficient production of •OH has been proven upon the oxygenation of FeS, but it is not clear whether the •OH produced can concurrently oxidize contaminants. Using As(III) as a representative of contaminants, we found that the oxygenation of 1 g/L FeS concurrently oxidized 95.9% of As(III) at an initial concentration of 1000 μg/L within 4 h (Figure 5). To evaluate the contribution of •OH to As(III) oxidation, 10 mM BA and 100 mM methanol were respectively added into the suspension for scavenging •OH. A remarkable decrease in As(III) oxidation was observed in the presence of both scavengers. As the influence of scavengers on the overall oxidation of Fe(II) was slight (SI Figure S8), the inhibition on As(III) oxidation confirmed the contribution of the •OH produced from FeS oxygenation. Different from the production of •OH and the decrease of total Fe(II), the lag time in the initial 0.5 h disappeared for the oxidation of As(III) (Figure 5). In this stage, the scavenging effects of BA and methanol on As(III) oxidation were negligible, indicating that the oxidation of As(III) proceeded on the solid surface. After 0.5 h, the scavenging effects of BA and methanol increased with the elapse of time, coinciding with the start of •OH

366 production. Previous investigations on photocatalytic oxidation of As(III) by TiO_2
367 documented that adsorbed •OH and O_2•$^-$ both contribute to the oxidation of adsorbed
368 As(III).[53–56] In the initial stage, the concentrations of FeS and intermediate sulfur
369 compounds are relatively high. As •OH was produced on FeS surface, the newly
370 formed adsorbed •OH may be largely consumed by FeS and adsorbed As(III) in the
371 initial stage before it escaped into the solution. The gradual consumption of FeS
372 decreased its reaction with •OH, rendering more •OH in the solution. Consequently,
373 the inhibitory effect with addition of scavengers increased with the progress of
374 oxygenation. The small portion of O_2•$^-$ which was produced from oxygenation of the
375 Fe(II) fed by Fe(III) reduction may also contribute to As(III) oxidation to some
376 extent.[55,56]

377 The results herein provide direct evidence for the involvement of •OH in the
378 concurrent As(III) oxidation with FeS oxygenation. Although oxidation of inorganic
379 contaminants coupled with FeS oxygenation has been previously noted by several
380 researchers, different reactive oxidants have been proposed.[11–14] According to our
381 finding, the contribution of •OH to contaminant oxidation depends on the
382 experimental conditions for FeS oxygenation, particularly the co-existence of reduced
383 components including organic buffers. In most of previous investigations, tens of mM
384 organic buffers were used to control pH during the oxygenation of FeS.[11–14] The high
385 concentrations of organic buffers may greatly screen the oxidation of low levels of
386 contaminants by •OH due to the scavenging effect, so the observed contaminant
387 oxidation in literature may be mainly due to other reactive oxidants.[11–14] In a recent

study conducted by Bi et al., the oxidative dissolution of nanocrystalline U(IV) coupled with FeS oxygenation was compared in the presence and absence of carbonate.[14] The suspension pH at 7.0 was respectively controlled by bicarbonate and 10 mM 3-(N-morpholino) propanesulfonic acid (MOPS) in the presence and absence of carbonate.[14] Interestingly, the oxidative dissolution of nanocrystalline U(IV) was significant in the presence of carbonate but negligible in the presence of 10 mM MOPS. Although the authors attributed the difference of oxidative dissolution to the effect of carbonate, we suspect the involvement of •OH could be one reason. In addition to the organic buffers, the other co-existing reduced components including FeS itself may also compete with contaminants for •OH during FeS oxygenation. This could be a reason for the overlook of oxidizing impact induced by FeS at oxic condition as well as the use of high dosages of FeS as a redox buffer for inhibiting oxidative mobilization of U(IV) and Tc(IV).[10,11] However, when FeS is at a low content or is gradually depleted, the oxidizing impact could presumably become significant, which was reflected by the oxidative mobilization of Tc(IV) sequestrated on sulfidated nZVI at a low S/Fe ratio or in the latter stage of oxygenation.[11]

Implications. In this study, production of •OH was confirmed from the oxygenation of FeS. The efficiency of FeS on •OH production was much higher than the other forms of reduced iron including siderite, pyrite and Fe^0 nanoparticles. Structural Fe(II) in FeS surface donated two electrons to O_2 with generation of H_2O_2, which was then decomposed by FeS to •OH. The •OH produced from FeS oxygenation can induce the concurrent oxidation of As(III). Although FeS has been

extensively tested for reducing contaminants under anoxic conditions,[2] the oxidizing effect under oxic conditions has been largely underestimated or even overlooked. As the intrusion of air into the anoxic FeS environments often happens, i.e., in the treatment of U,[14] contaminant oxidation could happen due to the •OH produced from FeS oxygenation, particularly at a low level of FeS content. It should be also cautious at the later stage when FeS is used as the redox buffer for preventing the oxidative mobilization of toxic metals. Aqueous FeS was observed in the field under anoxic conditions, and frequent oxygenation of the aqueous FeS in the oxic/anoxic interface has been substantiated.[46,47] FeS minerals produced from biological sulfate reduction are generally in the form of colloids,[13] rendering a strong mobility in the subsurface porous media.[13] FeS is capable of sequestrating many contaminants such as As, Hg and U.[2] As a consequence, transport of FeS colloids carrying contaminants from anoxic to oxic environments may suffer from oxygenation, which could produce •OH for the oxidative transformation of the carried contaminants. More investigations are needed to evaluate the oxidizing impact induced by FeS at oxic condition on contaminant transformation.

Supporting Information Available

Additional information: Figure S1–S8, XRD and XPS patterns for the pristine FeS particles, production of •OH upon oxygenation of different reduced iron, the linear dependence between •OH concentrations and FeS dosages, instantaneous concentrations of dissolved Fe^{2+} and $S_2O_3^{2-}$, variation of DO concentration,

production of •OH from H_2O_2 and FeS, decrease in total Fe(II) concentration during As(III) oxidation; Table S1, comparison of total Fe(II) oxidized and elemental S produced. This material is available free of charge via the Internet at http://pubs.acs.org.

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609

610 **Figure captions**

611 **Figure 1. C**umulative concentrations of •OH produced from oxygenation of FeS. The
612 dosage of FeS was 1 g/L.

613 **Figure 2.** (a) Cumulative concentrations of •OH, (b) total concentrations of Fe(II), (c)
614 production of sulfur compounds and (d) linear dependence of •OH production on total
615 Fe(II) content upon oxygenation of FeS at different dosages. Note the dosage of FeS
616 in (c) was 1 g/L FeS.

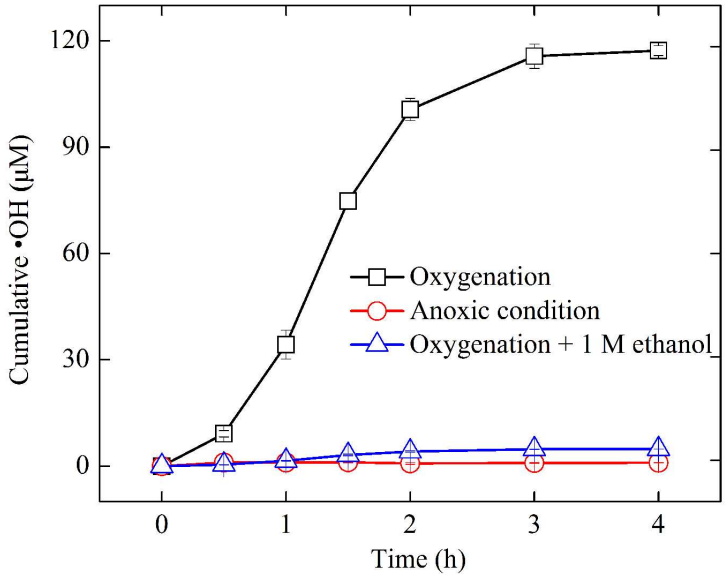
617 **Figure 3.** (a) XRD patterns and (b) k^3-weighted Fe K-edge EXAFS spectrums of the
618 minerals collected during the oxygenation of 3 g/L FeS at pH 7. Oxidation time is
619 indicated inside. Solid lines represent experimental data and dashed lines the best fit
620 for EXAFS spectrum.

621 **Figure 4.** (a) Production of H_2O_2 upon oxygenating different dosages of FeS, effects
622 of BPY, NBT and SOD on (b) H_2O_2 and (c) •OH production upon oxygenating 1 g/L
623 FeS. The concentrations of BPY, NBT and SOD were 0.5 mM, 1 mM and 60 U/L,
624 respectively.

625 **Figure 5.** Oxidation of As(III) by the •OH produced from oxygenation of 1 g/L FeS.
626 Note that the concentrations of As refer to As(III) and the total As in both aqueous and
627 solid phase.

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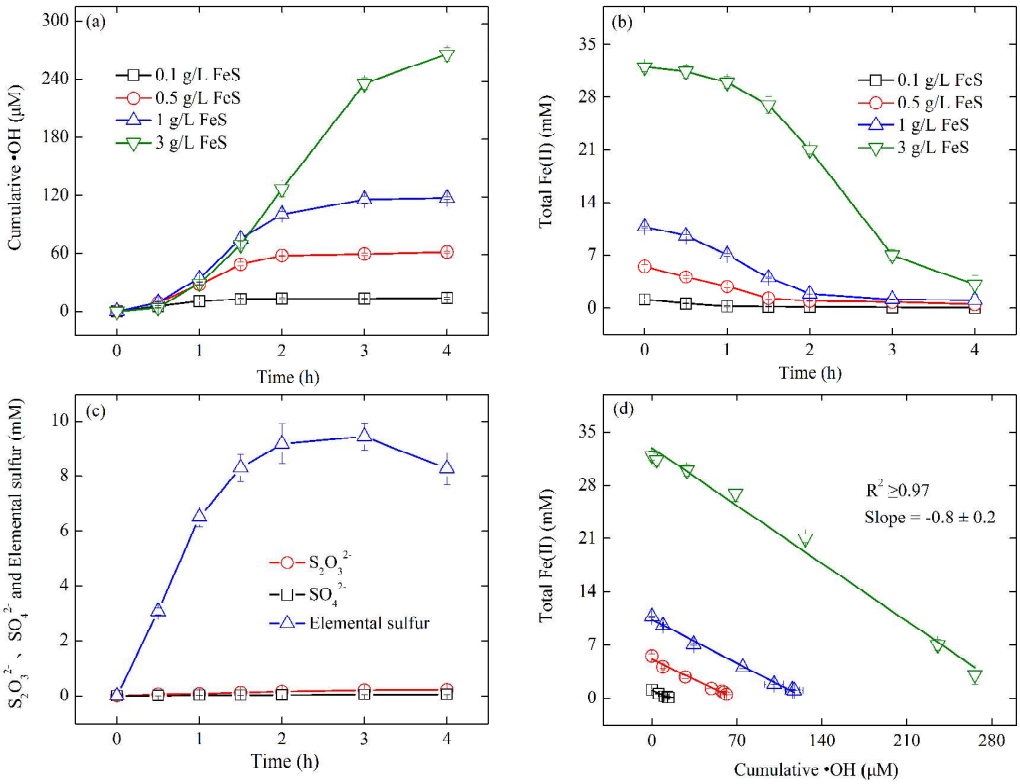


630

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634

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638 dosage of FeS in (c) was 1 g/L FeS and the data points in (d) were from (a) and (b).

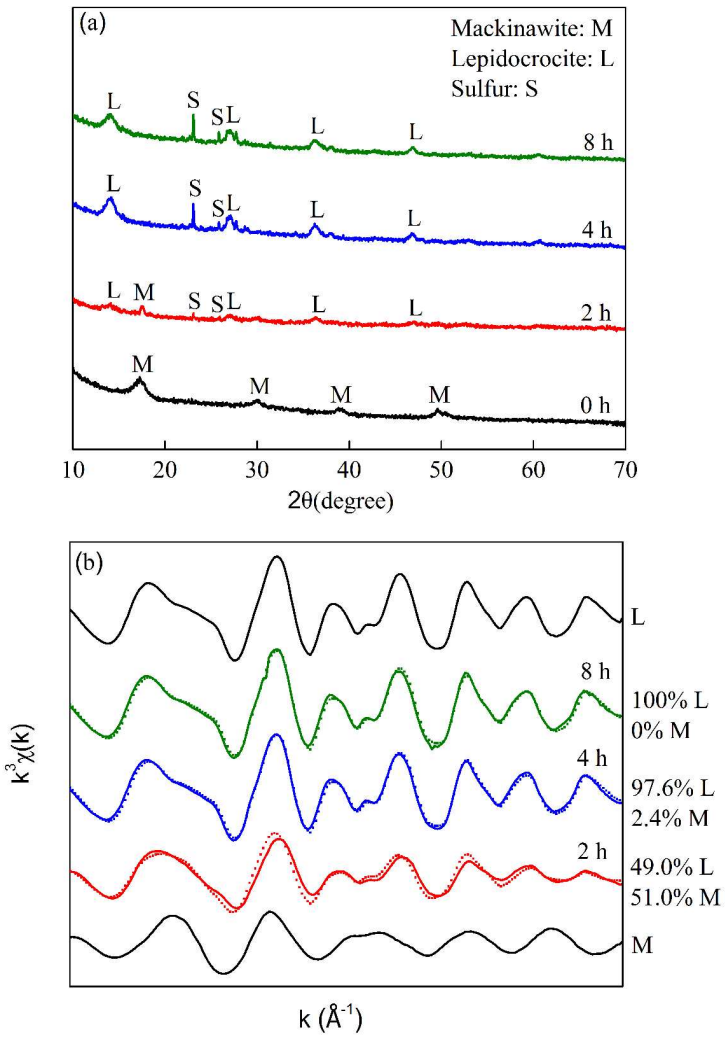


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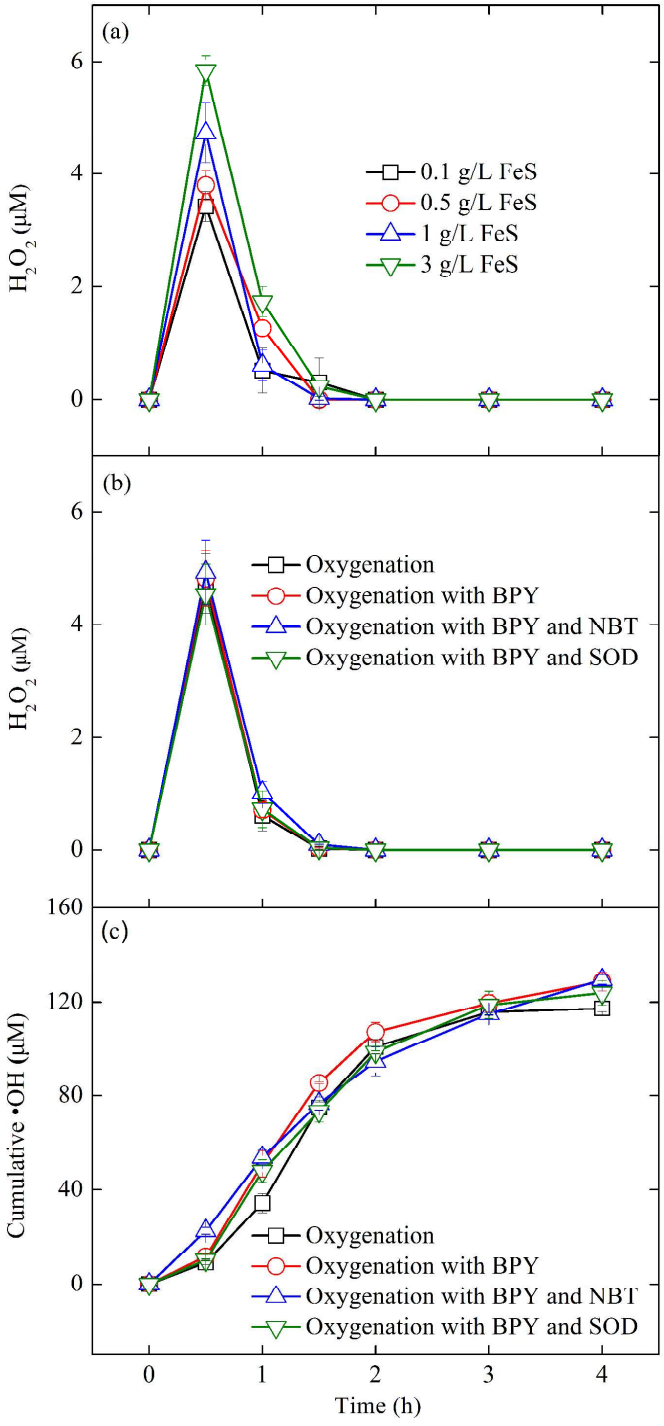
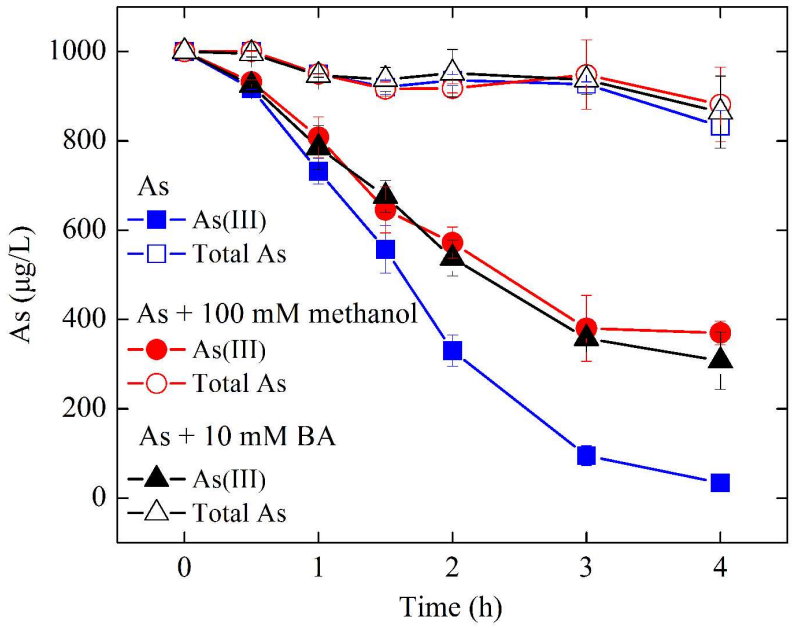


Figure 4. (a) Production of H_2O_2 upon oxygenating different dosages of FeS, effects of BPY, NBT and SOD on (b) H_2O_2 and (c) •OH production upon oxygenating 1 g/L FeS. The concentrations of BPY, NBT and SOD were 0.5 mM, 1 mM and 60 U/L,

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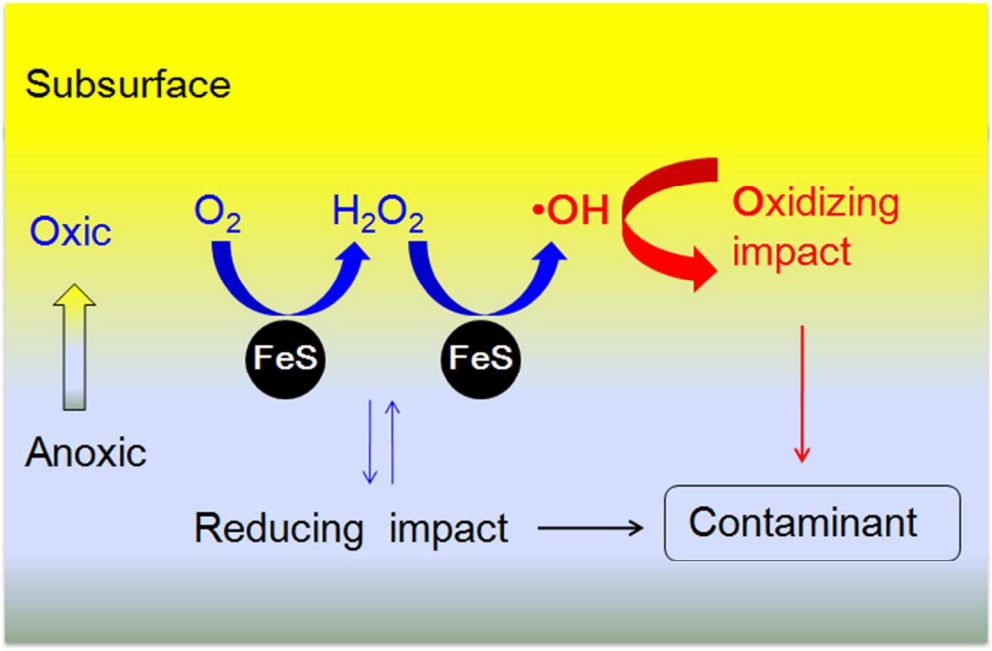


653

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